toys10_04-beat-runner

Beat Runner: manufacturing manual

> **Status: design only; nothing built.**

1. What is built

1. Notes spawn at the far end and scroll toward LED 0, getting faster every 40 steps (400 ms → 150 ms).
2. Press when a note is on LED 0 (perfect, 3 points) or LED 1 (good, 1 point). Combos multiply score; 5 misses end the game.
3. Two players alternate; highest score wins.

2. Deliverables

Item	Where
PCB (150 x 46 mm, 2 x 74HC595, 16 LEDs)	`utility10/pcb/release/beatrunner/` (released)
Firmware for the board	`toys10/electronic/release/pcb/beat_runner_rev_b.hex` (buzzer on D9, build `make pcb`)
Engine + tests	`toycore.c` (`br_*`)

3. Bill of materials

3a. Board parts

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
U1	ATmega328P-AU	TQFP-32_7x7mm_P0.8mm	ATMEGA328P-AU	C14877	SMT (JLC)
C1	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
C2	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
C3	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
R7	10k	R_0603_1608Metric	0603 10k 1%	C25804	SMT (JLC)
J2	ISP	PinHeader_2x03_P2.54mm_Vertical	HCTL PZ254-2-03-Z-8.5 (ZHOURI C5116479 had only 340 in stock)	C2894964	hand/THT
J1	BATT 3xAAA	JST_PH_B2B-PH-K_1x02_P2.00mm_Vertical	JST B2B-PH-K-S(LF)(SN)	C131337	hand/THT
SW1	Power	SW_SPDT_PCM12	C&K PCM12SMTR	C221841	SMT (JLC)

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
Q2	AO3401A	SOT-23	AO3401A	C15127	SMT (JLC)
C4	10u	C_0805_2012Metric	0805 10uF X5R	C15850	SMT (JLC)
BZ1	Buzzer	Buzzer_12x9.5RM7.6	YUEXIN KC1206 16 ohm passive magnetic 12x9.5 mm (2048 Hz)	C781869	hand/THT
Q1	MMBT3904	SOT-23	MMBT3904	C20526	SMT (JLC)
D2	1N4148WS	D_SOD-323	1N4148WS	C2128	SMT (JLC)
R6	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)
U2	74HC595	SOIC-16_3.9x9.9mm_P1.27mm	74HC595D	C5947	SMT (JLC)
U3	74HC595	SOIC-16_3.9x9.9mm_P1.27mm	74HC595D	C5947	SMT (JLC)
C5	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
C6	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
J3	Arcade button	JST_PH_B2B-PH-K_1x02_P2.00mm_Vertical	JST B2B-PH-K-S(LF)(SN)	C131337	hand/THT
D10	Red	LED_0603_1608Metric	Hubei KENTO KT-0603R red (Everlight 19-217 was out of stock)	C2286	SMT (JLC)
R20	330	R_0603_1608Metric	UNI-ROYAL 0603WAF3300T5E	C23138	SMT (JLC)
D11	Red	LED_0603_1608Metric	Hubei KENTO KT-0603R red (Everlight 19-217 was out of stock)	C2286	SMT (JLC)
R21	330	R_0603_1608Metric	UNI-ROYAL 0603WAF3300T5E	C23138	SMT (JLC)
D12	Red	LED_0603_1608Metric	Hubei KENTO KT-0603R red (Everlight 19-217 was out of stock)	C2286	SMT (JLC)
R22	330	R_0603_1608Metric	UNI-ROYAL 0603WAF3300T5E	C23138	SMT (JLC)
D13	Red	LED_0603_1608Metric	Hubei KENTO KT-0603R red (Everlight 19-217 was out of stock)	C2286	SMT (JLC)
R23	330	R_0603_1608Metric	UNI-ROYAL 0603WAF3300T5E	C23138	SMT (JLC)
D14	Red	LED_0603_1608Metric	Hubei KENTO KT-0603R red (Everlight 19-217 was out of stock)	C2286	SMT (JLC)
R24	330	R_0603_1608Metric	UNI-ROYAL 0603WAF3300T5E	C23138	SMT (JLC)
D15	Red	LED_0603_1608Metric	Hubei KENTO KT-0603R red (Everlight 19-217 was out of stock)	C2286	SMT (JLC)
R25	330	R_0603_1608Metric	UNI-ROYAL 0603WAF3300T5E	C23138	SMT (JLC)
D16	Red	LED_0603_1608Metric	Hubei KENTO KT-0603R red (Everlight 19-217 was out of stock)	C2286	SMT (JLC)

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
R26	330	R_0603_1608Metric	UNI-ROYAL 0603WAF3300T5E	C23138	SMT (JLC)
D17	Red	LED_0603_1608Metric	Hubei KENTO KT-0603R red (Everlight 19-217 was out of stock)	C2286	SMT (JLC)
R27	330	R_0603_1608Metric	UNI-ROYAL 0603WAF3300T5E	C23138	SMT (JLC)
D18	Red	LED_0603_1608Metric	Hubei KENTO KT-0603R red (Everlight 19-217 was out of stock)	C2286	SMT (JLC)
R28	330	R_0603_1608Metric	UNI-ROYAL 0603WAF3300T5E	C23138	SMT (JLC)
D19	Red	LED_0603_1608Metric	Hubei KENTO KT-0603R red (Everlight 19-217 was out of stock)	C2286	SMT (JLC)
R29	330	R_0603_1608Metric	UNI-ROYAL 0603WAF3300T5E	C23138	SMT (JLC)
D20	Red	LED_0603_1608Metric	Hubei KENTO KT-0603R red (Everlight 19-217 was out of stock)	C2286	SMT (JLC)
R30	330	R_0603_1608Metric	UNI-ROYAL 0603WAF3300T5E	C23138	SMT (JLC)
D21	Red	LED_0603_1608Metric	Hubei KENTO KT-0603R red (Everlight 19-217 was out of stock)	C2286	SMT (JLC)
R31	330	R_0603_1608Metric	UNI-ROYAL 0603WAF3300T5E	C23138	SMT (JLC)
D22	Red	LED_0603_1608Metric	Hubei KENTO KT-0603R red (Everlight 19-217 was out of stock)	C2286	SMT (JLC)
R32	330	R_0603_1608Metric	UNI-ROYAL 0603WAF3300T5E	C23138	SMT (JLC)
D23	Red	LED_0603_1608Metric	Hubei KENTO KT-0603R red (Everlight 19-217 was out of stock)	C2286	SMT (JLC)
R33	330	R_0603_1608Metric	UNI-ROYAL 0603WAF3300T5E	C23138	SMT (JLC)
D24	Red	LED_0603_1608Metric	Hubei KENTO KT-0603R red (Everlight 19-217 was out of stock)	C2286	SMT (JLC)
R34	330	R_0603_1608Metric	UNI-ROYAL 0603WAF3300T5E	C23138	SMT (JLC)
D25	Red	LED_0603_1608Metric	Hubei KENTO KT-0603R red (Everlight 19-217 was out of stock)	C2286	SMT (JLC)
R35	330	R_0603_1608Metric	UNI-ROYAL 0603WAF3300T5E	C23138	SMT (JLC)

3b. Bought-in items

Item	Rs
MCU + crystal	55
PCB	35
2 × 74HC595	14
16 LEDs + resistor arrays	30
Arcade button	15

Item	Rs
Buzzer	12
Battery/USB power	30
Enclosure	60
Packaging + manual	30
Assembly + test	45

- One 30 mm arcade button (J3), 3 x AAA holder + switch, enclosure, 16 x red 0603 LEDs are on the board.

4. Make, program, test

1. Order PCB and SMT assembly per STANDARD_PROCEDURES A/B; solder THT parts (battery JST, switch, buzzer, button socket, ISP).
2. Fuses and flash: `avrdude -p m328p -c usbasp -U lfuse:w:0xE2:m -U hfuse:w:0xD9:m -U efuse:w:0xFD:m` then `-U flash:w:toys10/electronic/release/pcb/beat_runner_rev_b.hex:i`.
3. Pin map: data D11, shift clock D12, latch D13 (shared with the ISP header: the LEDs flicker while programming), button D2, buzzer D9.
4. Test: LEDs light in sequence as notes scroll; press when a note reaches LED 0: perfect = 3 points (high beep), good at LED 1; 5 misses end the game (all LEDs light); speed rises every 40 steps (400 ms to 150 ms).
5. Current: at most a few LEDs lit at once (74HC595 allows 70 mA total); idle current under 15 mA.

5. Packaging, compliance

Toys QCO / IS 9873 applicability and testing; age grade and small-parts warnings; Legal Metrology labelling; battery safety, BIS toy-electrics, e-waste EPR, noise level; no licensed names or logos.

6. Known limits

Host tests: scoring windows, a perfect player never loses over 500+ notes, an idle player loses after 5 misses, speed-up bounded to 150 ms.

Perceived value vs mobile games; keep price ≤ Rs 799.